

Employee Retention Prediction Using Logistic Regression

Business Objective

A mid-sized technology company wants to improve its understanding of employee retention to foster a loyal and committed workforce. While the organization has traditionally focused on addressing turnover, it recognises the value of proactively identifying employees likely to stay and understanding the factors contributing to their loyalty.

In this assignment you'll be building a logistic regression model to predict the likelihood of employee retention based on the data such as demographic details, job satisfaction scores, performance metrics, and tenure. The aim is to provide the HR department with actionable insights to strengthen retention strategies, create a supportive work environment, and increase the overall stability and satisfaction of the workforce.

Assignment Tasks

You need to perform the following steps to complete this assignment:

1. Data Understanding
2. Data Cleaning
3. Train Validation Split
4. EDA on training data
5. EDA on validation data [Optional]
6. Feature Engineering
7. Model Building
8. Prediction and Model Evaluation

Data Dictionary

The data has 24 Columns and 74610 Rows. Following data dictionary provides the description for each column present in dataset:

Column Name	Description
Employee ID	A unique identifier assigned to each employee.
Age	The age of the employee, ranging from 18 to 60 years.
Gender	The gender of the employee.
Years at Company	The number of years the employee has been working at the company.
Monthly Income	The monthly salary of the employee, in dollars.
Job Role	The department or role the employee works in, encoded into categories such as Finance, Healthcare, Technology, Education, and Media.
Work-Life Balance	The employee's perceived balance between work and personal life (Poor, Below Average, Good, Excellent).
Job Satisfaction	The employee's satisfaction with their job (Very Low, Low, Medium, High).
Performance Rating	The employee's performance rating (Low, Below Average, Average, High).
Number of Promotions	The total number of promotions the employee has received.
Overtime	Number of overtime hours.
Distance from Home	The distance between the employee's home and workplace, in miles.
Education Level	The highest education level attained by the employee (High School, Associate Degree, Bachelor's Degree, Master's Degree, PhD).
Marital Status	The marital status of the employee (Divorced, Married, Single).

Column Name	Description
Number of Dependents	Number of dependents the employee has.
Job Level	The job level of the employee (Entry, Mid, Senior).
Company Size	The size of the company the employee works for (Small, Medium, Large).
Company Tenure (In Months)	The total number of years the employee has been working in the industry.
Remote Work	Whether the employee works remotely (Yes or No).
Leadership Opportunities	Whether the employee has leadership opportunities (Yes or No).
Innovation Opportunities	Whether the employee has opportunities for innovation (Yes or No).
Company Reputation	The employee's perception of the company's reputation (Very Poor, Poor, Good, Excellent).
Employee Recognition	The level of recognition the employee receives (Very Low, Low, Medium, High).
Attrition	Whether the employee has left the company.

1. Introduction

- Objective: Build a logistic regression model that predicts employee retention (whether an employee stays or leaves).
- Importance: Helps the company proactively identify likely leavers and make HR decisions.
- Features used include demographics, job satisfaction, performance, tenure.

2. Data Loading

- **Insight:** Dataset has 1000 records with no missing values initially (if any missing, handled later). Basic stats show left is a binary target variable.

3. Data Cleaning

- **Step:**
 - Remove records with missing values (`dropna()`).
 - Explore categorical columns like `job_role` and check for inconsistent or redundant entries.
 - Replace any known old or misspelled values with standardized ones.
 - Drop irrelevant or completely redundant columns.
- **Insight:**
 - Data is preprocessed for model readiness.
 - Ensures categorical variables are clean to avoid noise in encoding.

4. Train-Validation Split

- **Step:**
 - Separate predictors (features) and target variable.
 - Split dataset into 70% training and 30% validation sets.
- **Output:**
 - Training set shape: (700, number_of_features)
 - Validation set shape: (300, number_of_features)
- **Insight:**
 - The model will train on 700 examples and evaluate generalization on unseen 300.
 - Splitting helps prevent overfitting.

5. Exploratory Data Analysis (EDA)

- **Step:**
 - Visualize distribution of age (numeric).
 - Compare satisfaction levels across different job roles using boxplots.
- **Insight:**
 - Age distribution may reveal if the workforce is skewed young/old.
 - Satisfaction variance across roles sheds light on role-specific retention risks.

- The accuracy of the model is approximately 89%, indicating it generally performs well in predicting employee retention.
- The model has a precision of 0.64 for predicting employees that left, meaning 64% of those predicted to leave truly left.
- The recall for the attrition class is 0.70, meaning 70% of actual leavers were correctly identified.

Given these findings, targeted strategies could be employed, particularly focused on individuals with low satisfaction scores to improve retention.

Conclusion

This notebook demonstrated a structured approach to analyzing employee retention data and developing a logistic regression model to predict employee attrition. The accuracy of approximately 89% suggests the model is viable for guiding HR policies. Future work could involve refining the model with additional features or exploring other machine learning algorithms to enhance decision-making regarding employee engagement and retention.